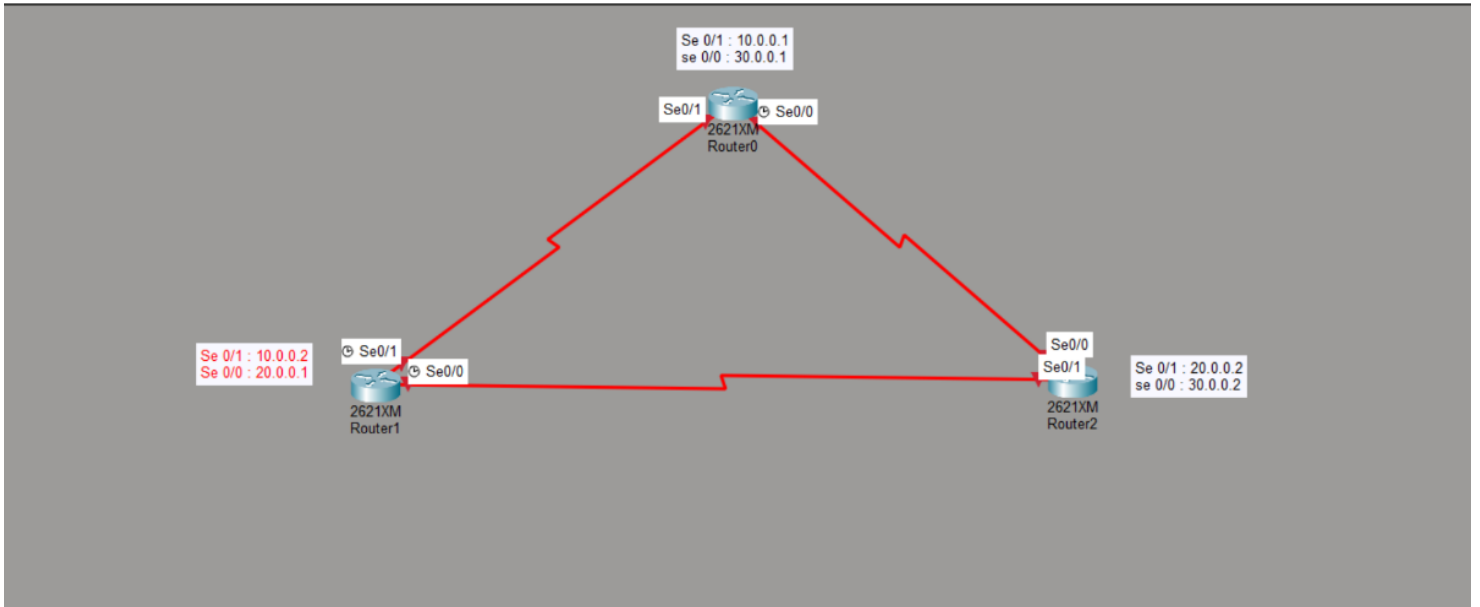
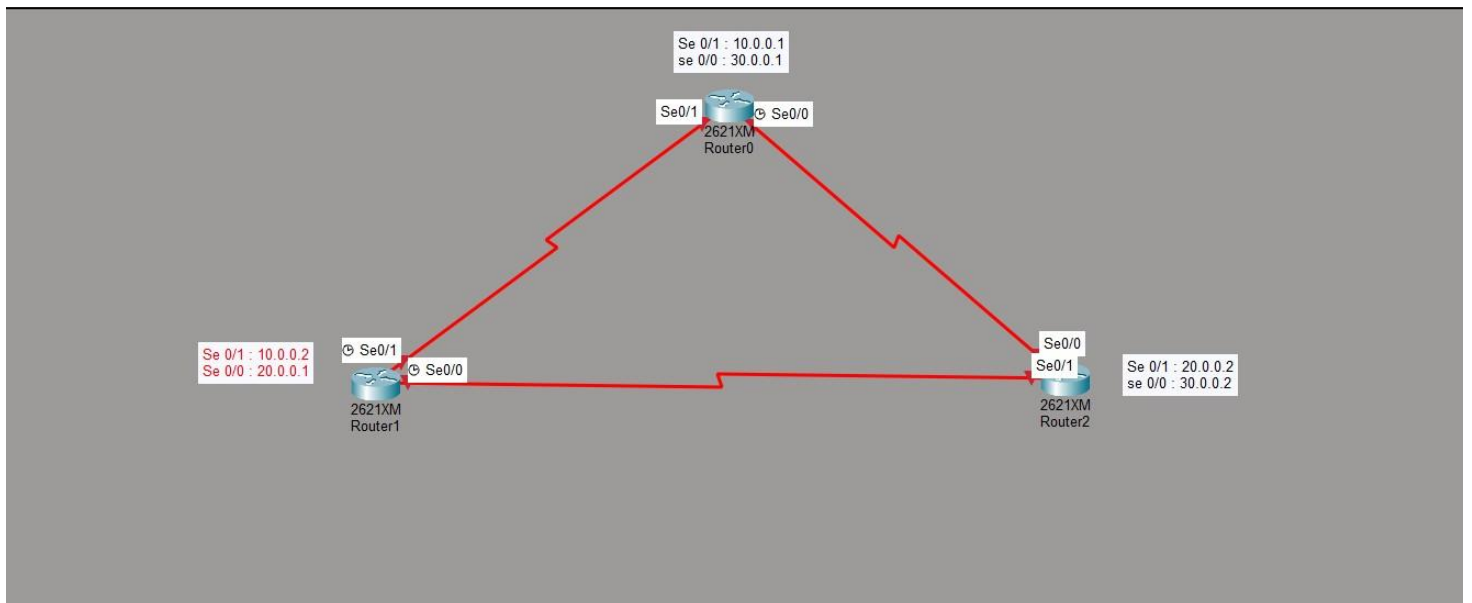


Task-1



Task 2 -Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.



Router 0

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/1
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0
Router(config-if)#ip address 30.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up

%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router(config-if)#exit
Router(config)#show ip in
Router(config)#show ip interface brief
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip
Router#show ip in
Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          unassigned      YES unset  administratively down  down
FastEthernet0/1          unassigned      YES unset  administratively down  down
Serial0/0                 30.0.0.1        YES manual  up                up
Serial0/1                 10.0.0.2        YES manual  up                up
Router#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/5/21 ms

Router#

```

Router 1

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/1
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          unassigned      YES unset  administratively down  down
FastEthernet0/1          unassigned      YES unset  administratively down  down
Serial0/0                 20.0.0.1        YES manual  up                up
Serial0/1                 10.0.0.1        YES manual  up                up
Router#ping 30.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping 20.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/12 ms

Router#

```

Router 2

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/0
Router(config-if)#ip address 30.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/1
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief

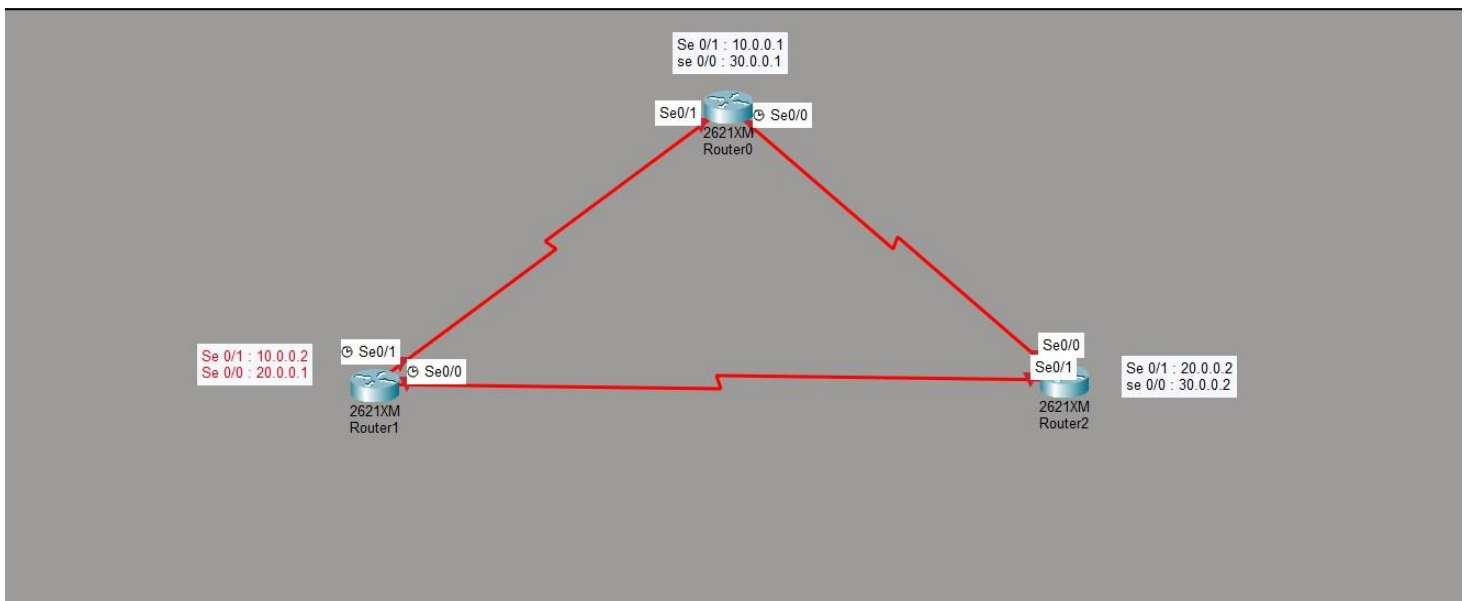

| Interface       | IP-Address | OK? | Method | Status                | Protocol |
|-----------------|------------|-----|--------|-----------------------|----------|
| FastEthernet0/0 | unassigned | YES | unset  | administratively down | down     |
| FastEthernet0/1 | unassigned | YES | unset  | administratively down | down     |
| Serial0/0       | 30.0.0.2   | YES | manual | up                    | up       |
| Serial0/1       | 20.0.0.2   | YES | manual | up                    | up       |


Router#ping 30.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/17 ms

Router#
```

Task-3

Q-Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks.



Router 0

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#show ip route
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Serial0/1
C    30.0.0.0/8 is directly connected, Serial0/0

Router#

```

Router 1

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
00:17:58: %OSPF-5-ADJCHG: Process 10, Nbr 30.0.0.1 on Serial0/1 from LOADING to FULL, Loading Done

Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Serial0/1
C    20.0.0.0/8 is directly connected, Serial0/0
O    30.0.0.0/8 [110/128] via 10.0.0.2, 00:00:09, Serial0/1

Router#

```

Router 2

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#
00:18:32: %OSPF-5-ADJCHG: Process 10, Nbr 30.0.0.1 on Serial0/0 from LOADING to FULL, Loading Done
00:18:32: %OSPF-5-ADJCHG: Process 10, Nbr 20.0.0.1 on Serial0/1 from LOADING to FULL, Loading Done

Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

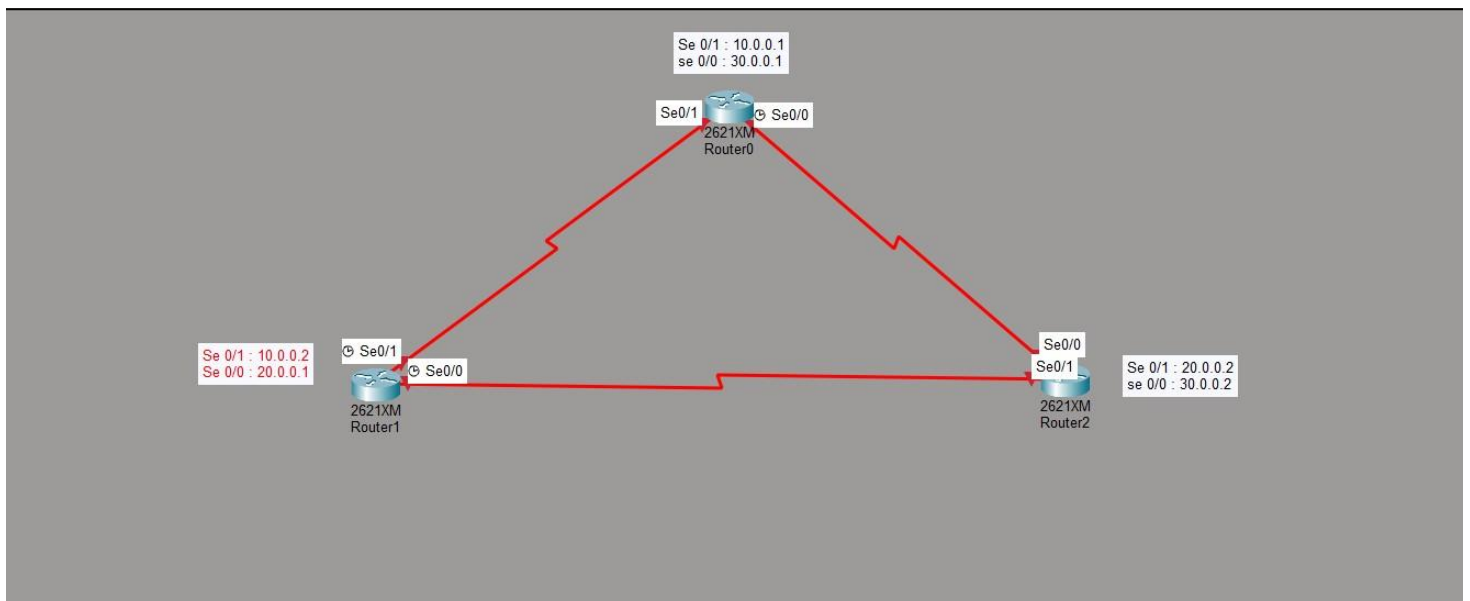
O    10.0.0.0/8 [110/128] via 30.0.0.1, 00:00:13, Serial0/0
      [110/128] via 20.0.0.1, 00:00:13, Serial0/1
C    20.0.0.0/8 is directly connected, Serial0/1
C    30.0.0.0/8 is directly connected, Serial0/0
Router#

```

Task- 4

Q-Observe and document the OSPF adjacency formation process by checking the output of the `show ip ospf neighbor` command on each router.

Hint: Look for the FULL/DR or FULL/BDR state to confirm adjacency.



Router 0

```

Router#en
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
20.0.0.1       0     FULL/-          00:00:39    10.0.0.1       Serial0/1
30.0.0.2       0     FULL/-          00:00:32    30.0.0.2       Serial0/0
Router#

```

Router 1

```
Router#en
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
30.0.0.2	0	FULL/ -	00:00:33	20.0.0.2	Serial0/0
30.0.0.1	0	FULL/ -	00:00:39	10.0.0.2	Serial0/1

```
Router#
```

Router 2

```
Router#en
Router#show ip ospf neighbor
```

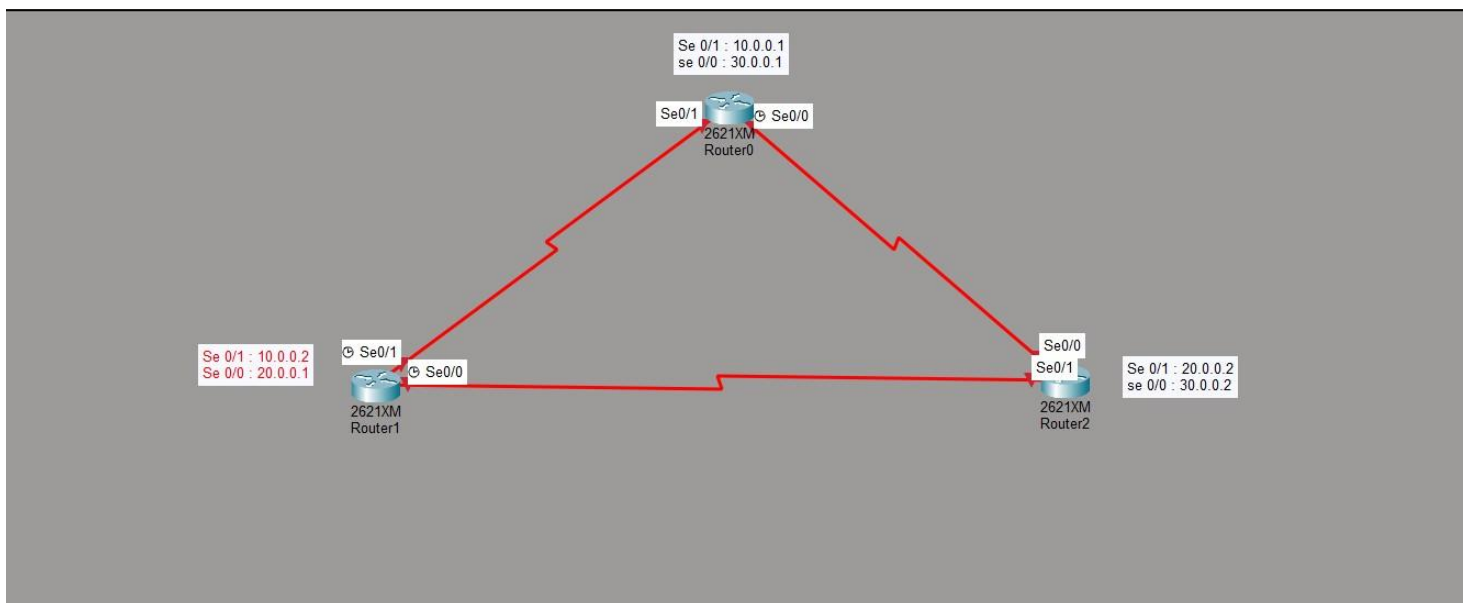
Neighbor ID	Pri	State	Dead Time	Address	Interface
20.0.0.1	0	FULL/ -	00:00:38	20.0.0.1	Serial0/1
30.0.0.1	0	FULL/ -	00:00:38	30.0.0.1	Serial0/0

```
Router#
```

Task- 5

Q-Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router.

Hint: Look for the FULL/DR or FULL/BDR state to confirm adjacency.



Router 0

```
Router>
Router>en
Router#show ip route ospf
```

O	20.0.0.0	[110/128]	via 10.0.0.1	00:06:47	Serial0/1
			[110/128]	via 30.0.0.2	00:06:47, Serial0/0

```
Router#
```

Router 1

```
Router#
Router#show ip route ospf
O    30.0.0.0 [110/128] via 10.0.0.2, 00:07:10, Serial0/1
      [110/128] via 20.0.0.2, 00:07:10, Serial0/0

Router#
```

Router 2

```
Router>
Router>en
Router#show ip route ospf
O    10.0.0.0 [110/128] via 30.0.0.1, 00:08:00, Serial0/0
      [110/128] via 20.0.0.1, 00:08:00, Serial0/1

Router#
```